

Tugas Week 10
Jaringan Komputer



Nama Dan Nim:
Fathan Aditya Rachman, 103072400153

Tujuan : Mahasiswa dapat menginvestigasi cara kerja protokol IP menggunakan Wireshark

Menangkap paket dari eksekusi traceroute >tracert gaia.cs.umass.edu

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz>tracert gaia.cs.umass.edu

Tracing route to gaia.cs.umass.edu [128.119.245.12]
over a maximum of 30 hops:

  1  16 ms  3 ms  3 ms  10.218.0.253
  2  19 ms  14 ms  10 ms  10.217.15.254
  3  25 ms  7 ms  4 ms  36.67.147.33
  4  34 ms  26 ms  *  180.240.190.101
  5  *  *  *  Request timed out.
  6  41 ms  24 ms  24 ms  180.240.204.8
  7  60 ms  26 ms  26 ms  be4126.ccr81.sin01.atlas.cogentco.com [154.18.3.249]
  8  197 ms  193 ms  191 ms  be9040.ccr41.lax04.atlas.cogentco.com [154.54.87.145]
  9  191 ms  192 ms  200 ms  be3271.ccr41.lax01.atlas.cogentco.com [154.54.42.101]
 10  204 ms  204 ms  203 ms  be2931.ccr31.phx01.atlas.cogentco.com [154.54.44.85]
 11  208 ms  209 ms  211 ms  be5471.ccr21.elp02.atlas.cogentco.com [154.54.166.57]
 12  222 ms  222 ms  221 ms  be3821.ccr31.dfw01.atlas.cogentco.com [154.54.165.25]
 13  236 ms  240 ms  235 ms  be5383.ccr31.oma02.atlas.cogentco.com [154.54.7.234]
 14  242 ms  243 ms  243 ms  be5214.ccr41.ord01.atlas.cogentco.com [154.54.165.134]
 15  249 ms  247 ms  249 ms  port-channel2717.ccr91.cle04.atlas.cogentco.com [154.54.6.222]
 16  250 ms  251 ms  252 ms  be0822.ccr21.alb02.atlas.cogentco.com [154.54.169.186]
 17  267 ms  259 ms  261 ms  be8077.rcr71.orh02.atlas.cogentco.com [154.54.160.213]
 18  262 ms  262 ms  264 ms  be2191.rcr71.orh01.atlas.cogentco.com [154.54.160.70]
 19  262 ms  264 ms  261 ms  38.32.33.83
 20  275 ms  262 ms  262 ms  69.16.0.9
 21  276 ms  261 ms  266 ms  69.16.3.0
 22  280 ms  265 ms  265 ms  core2-rt-et-4-3-0.gw.umass.edu [192.80.83.105]
 23  275 ms  272 ms  271 ms  n5-rt-1-1-et-10-0-0.gw.umass.edu [128.119.0.10]
 24  268 ms  269 ms  268 ms  128.119.77.235
 25  *  *  *  Request timed out.
 26  266 ms  267 ms  267 ms  cics-rt-xe-0-0-0.gw.umass.edu [128.119.3.32]
 27  *  *  *  Request timed out.
 28  266 ms  270 ms  265 ms  gaia.cs.umass.edu [128.119.245.12]

Trace complete.

C:\Users\fairuuz>
```

```
*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src == 10.218.10.7 && ip.dst == 128.119.245.12

No. Time Source Destination Protocol Length Info
2132 79.6305360.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=62/15872, ttl=1 (no response found!)
2133 79.6466705.. 10.218.0.253 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2134 79.6494364.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=63/16128, ttl=1 (no response found!)
2135 79.6525984.. 10.218.0.253 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2136 79.6553062.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=64/16384, ttl=1 (no response found!)
2137 79.6584812.. 10.218.0.253 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2142 85.6231090.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=65/16640, ttl=2 (no response found!)
2143 85.6422256.. 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2144 85.6432761.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=66/16896, ttl=2 (no response found!)
2145 85.6580224.. 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2146 85.6584230.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=67/17152, ttl=2 (no response found!)
2147 85.6684418.. 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2568 91.4050117.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=68/17408, ttl=3 (no response found!)
2569 91.4301371.. 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2570 91.4334573.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=69/17664, ttl=3 (no response found!)
2571 91.4406517.. 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2572 91.5399319.. 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=70/17920, ttl=3 (no response found!)
2573 91.5447210.. 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)

Frame 2132: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0
Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: RuijieNetwork cf:9d:26 (4
Internet Protocol Version 4, Src: 10.218.10.7, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 92
Identification: 0x99ee (39406)
0000 .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
Protocol: ICMP (1)
Header Checksum: 0x954e [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.218.10.7
Destination Address: 128.119.245.12
[Stream index: 14]
Internet Control Message Protocol

Packets: 6203 - Displayed: 155 (2.5%) - Dropped: 0 (0.0%) Profile: Default
```

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src == 10.218.10.7 && ip.dst == 128.119.245.12

No.	Time	Source	Destination	Protocol	Length	Info
2132	79.6305360	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=62/15872, ttl=1 (no response found!)
2133	79.6466705	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2134	79.6494364	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=63/16128, ttl=1 (no response found!)
2135	79.6523984	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2136	79.6553062	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=64/16384, ttl=1 (no response found!)
2137	79.6584837	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2342	85.6231090	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=65/16640, ttl=2 (no response found!)
2343	85.6422256	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2344	85.6432761	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=66/16896, ttl=2 (no response found!)
2345	85.6580224	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2346	85.6584230	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=67/17152, ttl=2 (no response found!)
2347	85.6684418	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2568	91.4050117	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=68/17408, ttl=3 (no response found!)
2569	91.4301371	36.67.147.33	10.218.10.7	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2570	91.4334573	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=69/17664, ttl=3 (no response found!)
2571	91.4406517	36.67.147.33	10.218.10.7	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2572	91.5399319	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=70/17920, ttl=3 (no response found!)
2573	91.5447210	36.67.147.33	10.218.10.7	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)

Frame 2133: Packet, 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface 0

Ethernet II, Src: RuijieNetwork_cf:9d:26 (48:81:d4:cf:9d:26), Dst: LiteonTechno_af:cf:7d (28:91:af:cf:7d:28)

Internet Protocol Version 4, Src: 10.218.0.253, Dst: 10.218.10.7

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)

Total Length: 120

Identification: 0xba35 (47669)

0000 = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x9ed8 [validation disabled]

[Header checksum status: Unverified]

Source Address: 10.218.0.253

Destination Address: 10.218.10.7

[Stream index: 15]

Internet Control Message Protocol

wireshark_Wi-FiQ2RW03.pcapng

Packets: 6203 · Displayed: 155 (2.5%) · Dropped: 0 (0.0%)

Profile: Default

IPv4 Dasar

TTL 1 :

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
2132	79.6305360	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=62/15872, ttl=1 (no response found!)
2133	79.6466705	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2134	79.6494364	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=63/16128, ttl=1 (no response found!)
2135	79.6523984	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2136	79.6553062	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=64/16384, ttl=1 (no response found!)
2137	79.6584837	10.218.0.253	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2155	80.0900239	10.218.0.253	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2201	81.6041055	10.218.0.253	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2258	83.1113684	10.218.0.253	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2342	85.6231090	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=65/16640, ttl=2 (no response found!)
2343	85.6422256	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2344	85.6432761	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=66/16896, ttl=2 (no response found!)
2345	85.6580224	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2346	85.6584230	10.218.10.7	128.119.245.12	ICMP	106	Echo (ping) request id=0x0001, seq=67/17152, ttl=2 (no response found!)
2347	85.6684418	10.217.15.254	10.218.10.7	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
2351	85.6957205	10.217.15.254	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2405	87.1884189	10.217.15.254	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)
2451	88.6899068	10.217.15.254	10.218.10.7	ICMP	120	Destination unreachable (Port unreachable)

Frame 2132: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0

Ethernet II, Src: LiteonTechno_af:cf:7d (28:b2:b9:af:cf:7d), Dst: RuijieNetwork_cf:9d:26 (48:81:d4:cf:9d:26)

Internet Protocol Version 4, Src: 10.218.10.7, Dst: 128.119.245.12

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 92

Identification: 0x99ee (39406)

0000 = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 1

Protocol: ICMP (1)

Header Checksum: 0x954e [validation disabled]

[Header checksum status: Unverified]

Source Address: 10.218.10.7

Destination Address: 128.119.245.12

[Stream index: 14]

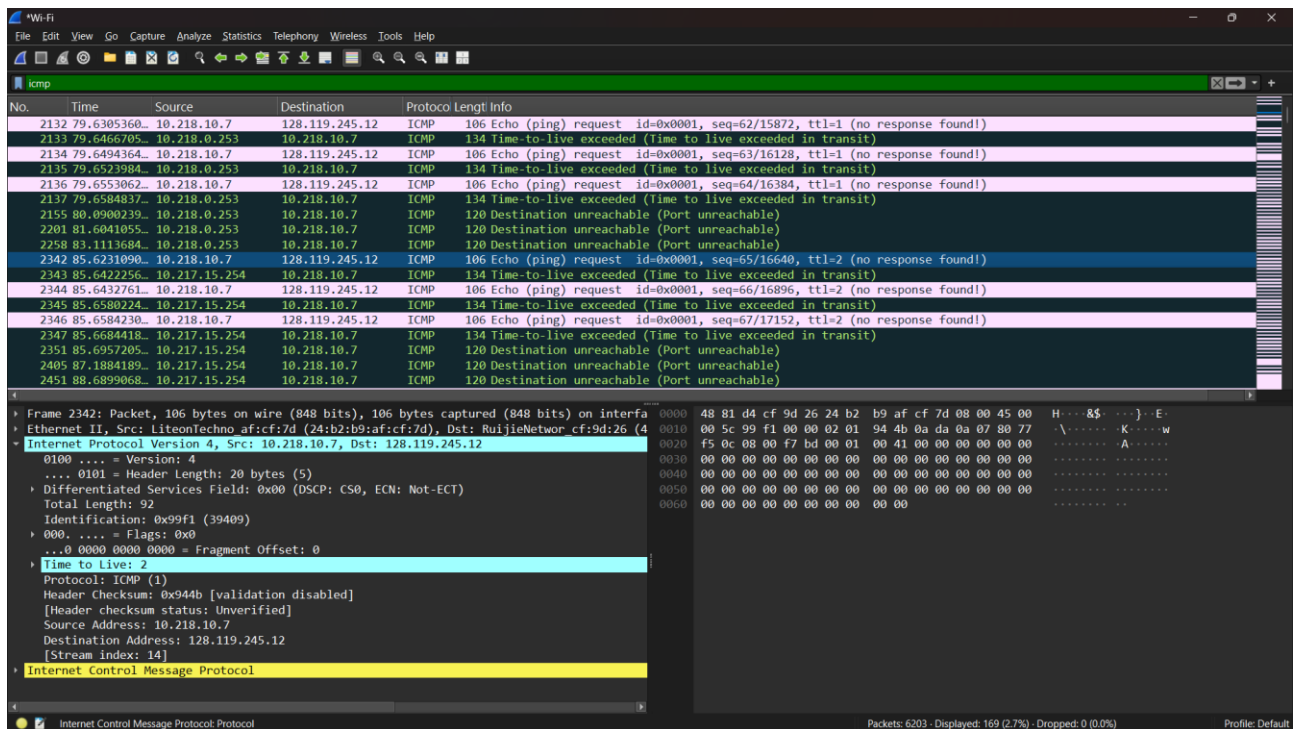
Internet Control Message Protocol

Internet Control Message Protocol: Protocol

Packets: 6203 · Displayed: 169 (2.7%) · Dropped: 0 (0.0%)

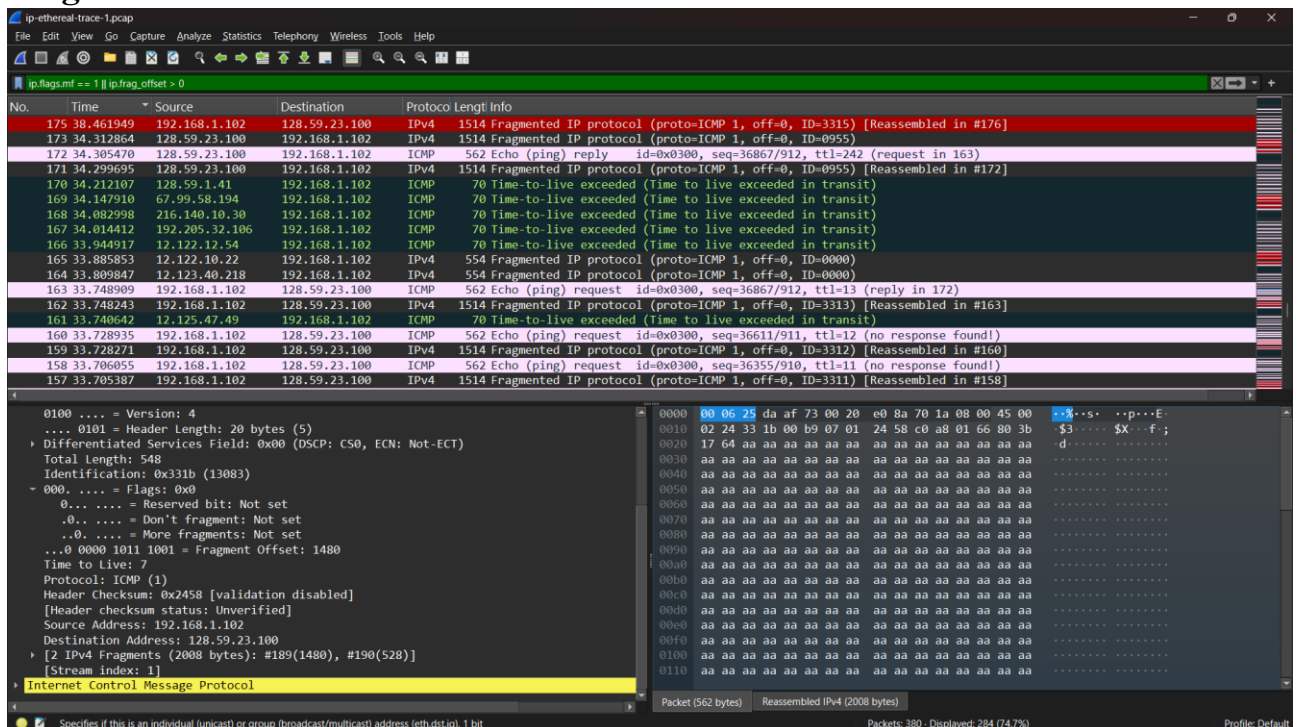
Profile: Default

TTL 2 :



Pada hasil capture Wireshark, terlihat bahwa komputer mengirim ICMP Echo Request dengan nilai TTL = 1 ke tujuan 128.119.245.12. Router pertama kemudian mengurangi nilai TTL menjadi 0 dan mengirimkan balasan ICMP “Time-To-Live Exceeded”. Hal ini menunjukkan bahwa traceroute berhasil mengidentifikasi hop pertama dalam jaringan.

Fregmentasi



Pada hasil analisis ditemukan fragmentasi IP. Hal ini ditunjukkan oleh adanya paket dengan nilai Fragment Offset lebih dari 0, yaitu sebesar 1480. Selain itu, nilai More Fragments (MF)

pada paket tersebut adalah 0, yang menunjukkan bahwa paket tersebut merupakan fragment terakhir. Fragmentasi terjadi karena datagram yang dikirim melebihi ukuran Maximum Transmission Unit (MTU), sehingga dipecah menjadi beberapa bagian. Hal ini diperkuat dengan adanya beberapa paket yang memiliki nilai Identification yang sama.

IPv6

Berdasarkan hasil analisis menggunakan Wireshark, ditemukan paket DNS yang berisi permintaan tipe A (IPv4), seperti pada domain youtube.com dan doubleclick.net. Namun, tidak ditemukan paket DNS dengan tipe AAAA (IPv6). Hal ini menunjukkan bahwa pada saat proses capture dilakukan, sistem hanya melakukan resolusi alamat IPv4 dan tidak melakukan permintaan alamat IPv6. Kemungkinan hal ini disebabkan oleh konfigurasi jaringan atau DNS server yang tidak mengutamakan penggunaan IPv6.

